

NEUROMATCH ACADEMY · COMPUTATIONAL NEUROSCIENCE 2026

Functional Connectivity **Reconfiguration** in N-back Working Memory

— HCP N-back · participant-level prediction

MEET THE GAMMAS



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Does load-related brain connectivity predict working-memory performance?

Functional connectivity — how strongly two brain regions rise and fall together. **Reconfiguration** — how that coupling changes as memory load rises.

WE PREDICTED TWO THINGS

01 PATTERN

A 78-feature FC fingerprint of the 2-back - 0-back change predicts performance.

02 DIRECTION

Higher load shifts networks toward integration; larger shifts accompany better performance.

ANCHOR · AVERY ET AL. 2020

Predicted 2-back accuracy from task FC in this HCP dataset — $r = 0.36$. We ask whether the **change between loads** carries that signal.

A distributed pattern and a one-number direction are different claims.

We evaluated a 78-feature FC difference in held-out people

System segregation = (mean within-network FC - mean between-network FC) ÷ mean within-network FC, per participant per condition (Chan et al. 2014).

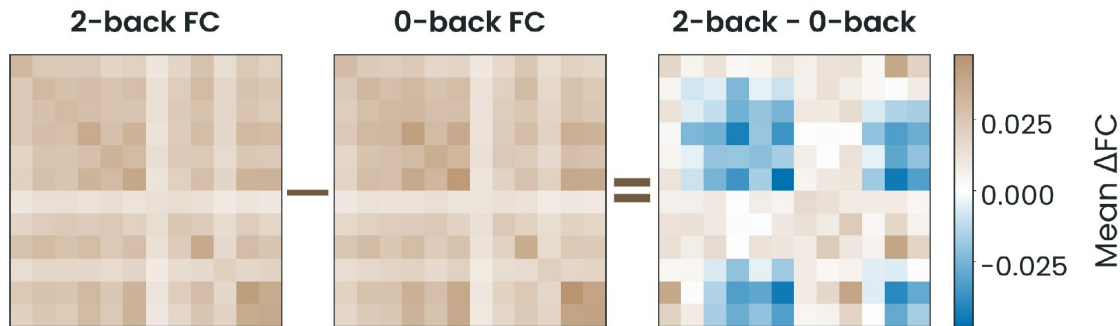
COHORT

01 336 participants HCP N-back task · 360 Glasser ROIs

CONDITION FRAMES

02 0-back and 2-back kept separate no temporal overlap

From frames to a 78-value fingerprint



Pearson FC between ROI time courses, summarized over 12 Cole-Anticevic networks · 12 within + 66 between = 78 values / person

MODEL & HELD-OUT EVALUATION

03 StandardScaler + RidgeCV · in-fold

Repeated 5-fold CV

Fixed holdout

1000-permutation null

B→A transfer · 301 → 100

All splits hold out participants · ± = split SD, not CI

RESULTS

The FC pattern predicted performance and transferred to a separate cohort

PRIMARY REPEATED CV

$r = 0.366 \pm 0.024$

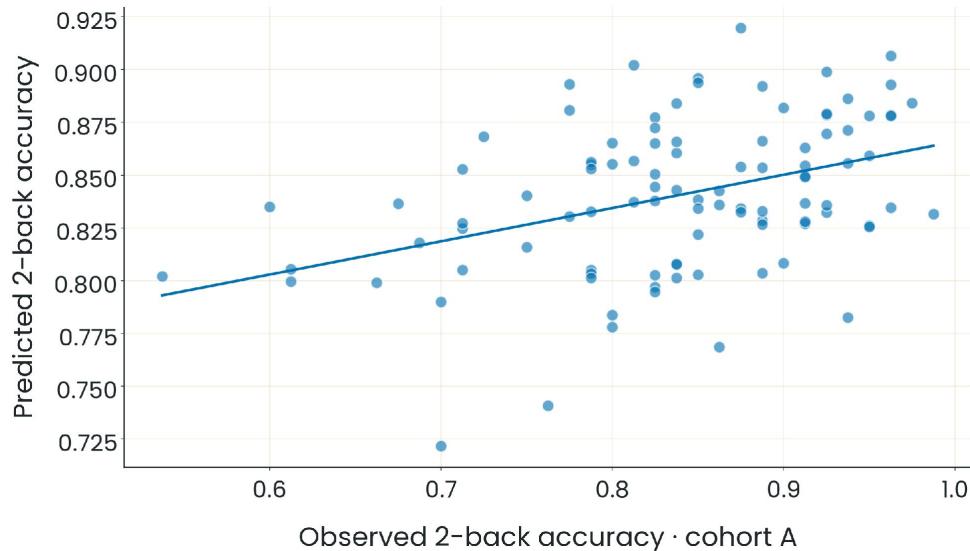
336 participants · 78 FC features

B → A TRANSFER

$r = 0.398$

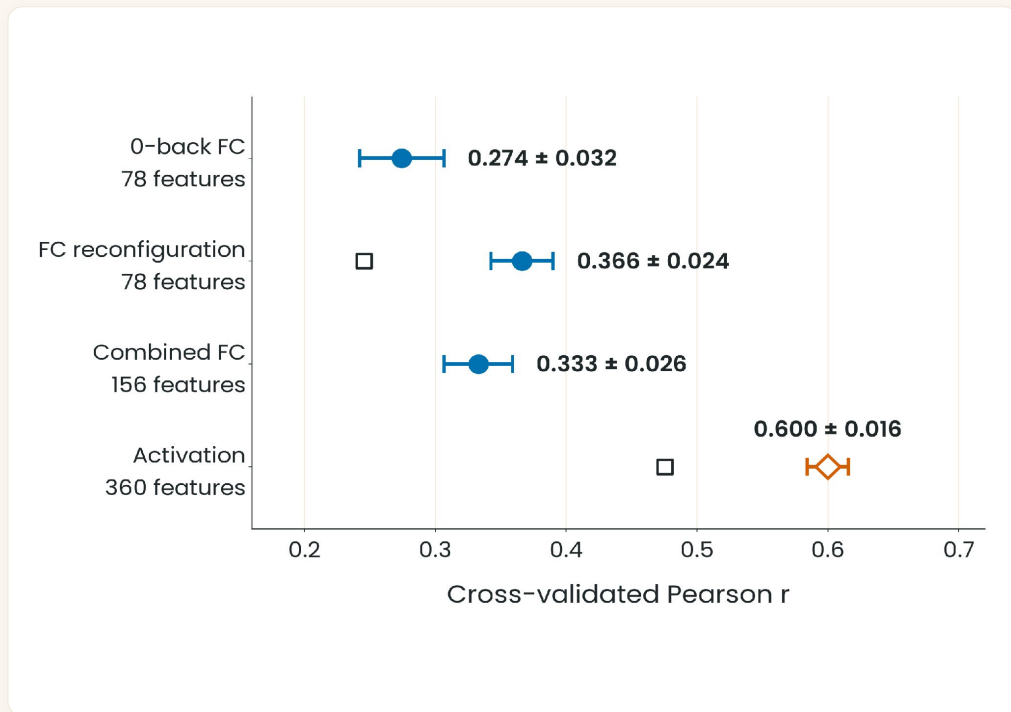
95% CI [0.25, 0.53] 301 B → 100 A · 35 shared identities removed

- Identity-disjoint same-HCP transfer — not independent-site validation.



Each dot = one participant. X: observed 2-back accuracy (cohort A). Y: accuracy predicted by a model that never saw them.

Two checks that narrowed what we can claim



X: cross-validated Pearson r · filled = full sample · open squares = held-out cross-run generalization.

Direction — predicted, then refined

Group level — real:

segregation fell under load

$0.3271 \rightarrow 0.3035 \cdot \Delta = -0.0236 \cdot p = 3.45 \times 10^{-5}$

Per person — weak:

larger shifts did not track performance

$r = -0.105 \cdot p = .054$ (n.s.)

CAVEATS

● A specificity check, not a contest

360 activation vs 78 FC features, post hoc; 0-back alone predicts as well (**0.571**).

→ not load-specific; FC reconfiguration stays the load-specific measure.

● CVR uncontrolled

Activation is a raw BOLD amplitude difference, not a GLM β ; this dataset carries no CVR proxy.

Predictive signal survives; connectivity-specific mechanism remains unresolved

Survives

the pattern hypothesis

- A **78-feature FC difference** predicts unseen 2-back accuracy.
- Model **transfers** across identity-disjoint HCP cohorts — $r = 0.398$.

Refined

the directional hypothesis

- **Segregation fell** under load — but larger shifts did not predict performance.
- **No clear gain** beyond 0-back FC.
- **No clear gain** over activation (unmatched comparison).

Unresolved

open questions

- Is the signal **connectivity-specific**, or shared with activation?
- **Vascular reactivity (CVR)** uncontrolled in the activation benchmark.

THE GAMMAS

Backup

Opened only on demand during Q&A.

Directional result

Validation detail

Future work

References & guardrails

Group direction was real; the individual link was weak

GROUP MEAN SEGREGATION

0.3271 → 0.3035

0-back → 2-back

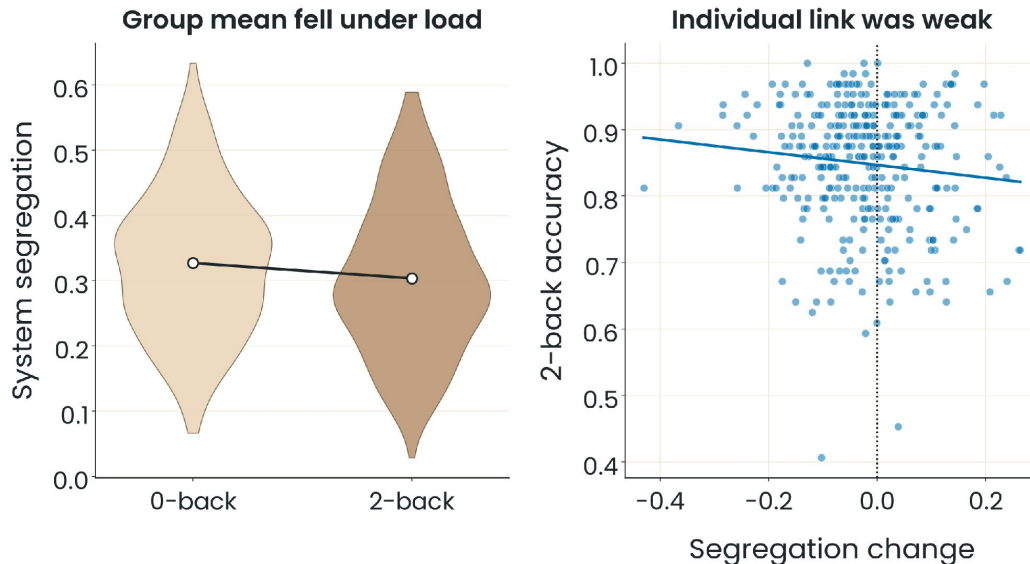
PAIRED CHANGE

$\Delta = -0.0236 \cdot p = 3.45 \times 10^{-5}$

ACROSS PARTICIPANTS

$r = -0.105 \cdot p = .054$

- A reliable mean shift does not establish individual predictive relevance.



Left: system segregation per condition. Right: 2-back accuracy vs each person's segregation change.

The checks behind the primary result

Fixed holdout · 67 unseen participants

$r = 0.312$

Full-refit null · seed 42

$r = 0.405 \cdot p \approx .001$

B→A A-label permutation

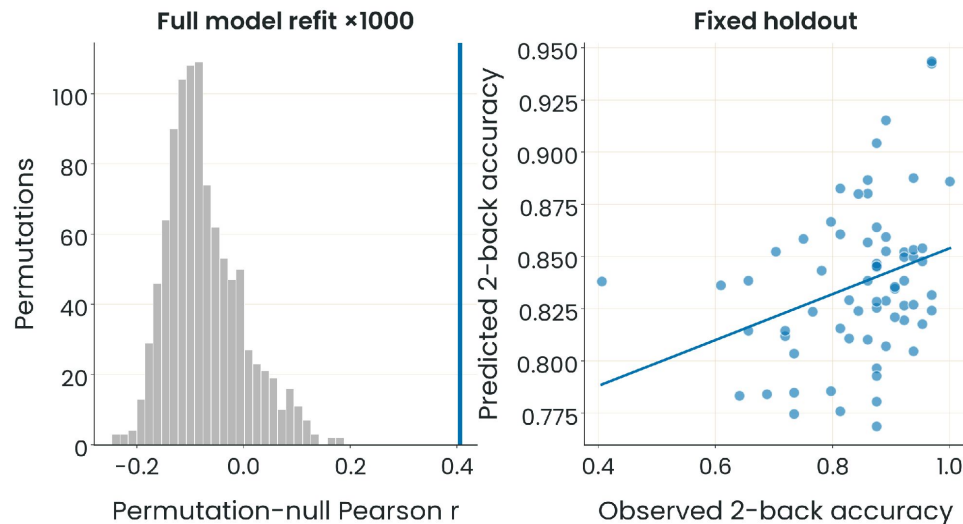
$p \approx .001$

Reconfiguration over 0-back FC **$\Delta R^2 = +0.034 \cdot \text{n.s.}$**

FC over activation

$\Delta R^2 = -0.003 \cdot \text{n.s.}$

- The permutation p belongs only to seed-42 $r = 0.405$; the holdout is a separate split.



Left: seed-42 refit sits far above the 1000-permutation null. Right: fixed-holdout predicted vs observed accuracy.

Five tests could resolve the remaining question

- 1 Match activation and FC dimensionality in nested participant-level CV.
- 2 Separate task coactivation from coupling with a prespecified estimator.
- 3 Test independent-site, repeat-session, family-aware generalization.
- 4 Run the activation model on resting-state amplitude — if rest predicts 2-back accuracy as well as task frames, the signal is task-independent.
- 5 Model the load transition as a dynamic graph (ST-GNN) instead of one static difference, so reorganisation is learned rather than summarized.

DECISION CRITERION FC must add reliable held-out value beyond activation and single-condition FC.

References and statistical guardrails

REFERENCES

Avery et al. (2020) · Distributed patterns of FC predict WM · J. Cognitive Neuroscience

Chan et al. (2014) · Decreased segregation of brain systems · PNAS

Murphy et al. (2020) · Multimodal network dynamics in WM · Nature Communications

Masharipov et al. (2024) · Task-modulated FC methods · Communications Biology

Hedge et al. (2018) · The reliability paradox · Behavior Research Methods

Logothetis (2008) · What we can and cannot do with fMRI · Nature

Full annotated bibliography · manuscript/references.md

STATISTICAL GUARDRAILS

$r = 0.366 \pm 0.024$

mean $\pm$ split SD across 20 repeated-CV partitions

seed-42 $r = 0.405 \cdot p \approx .001$

full-refit permutation (1/1001)

$B \rightarrow A$ $r = 0.398 \cdot CI [0.25, 0.53]$

identity-disjoint same-HCP transfer